90Ten GROUP

Smart Helmet: How v1.00 Works in the Existing Environment

How the application works while the helmets still use the supplier's existing servers.

Prepared for **Thella Group Services and Imperium Corporate** Designed and prepared by **90Ten Group** 20 October 2026 v1.00

01 What the existing environment is

v1.00 does not change the helmet. It uses what is already there and changes only where the helmet sends its reports. Message formats follow the supplier's Interface Document for Device-side Connecting.

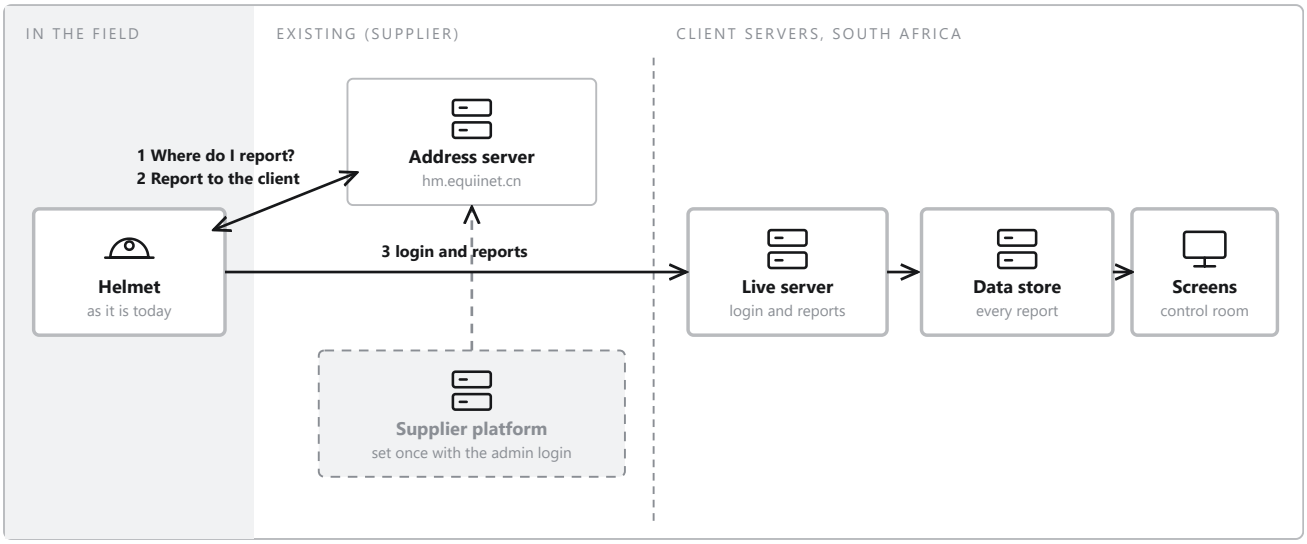
Part	Who has it	What we do with it
The helmet	The supplier made it. It has a SIM card for 4G.	Nothing. It stays as it is.
The address server	The supplier. It sits at hm.equiinet.cn and is built into the helmet.	Nothing. The helmet still asks it where to report.
The supplier platform	The supplier. We get an admin login for the helmets.	We log in once and change the two addresses to the client's server.
The mobile network	The SIM card on each helmet.	Nothing. The helmet reports over 4G on the surface.
The client's servers	The client's. They run in South Africa.	They receive the reports and keep them and show them.

WHAT CHANGES AND WHAT DOES NOT

What changes is the two addresses the helmet is told to report to. What does not change is the helmet itself and its software and the supplier address server it asks first.

02 How a helmet reaches the client's server

The helmet first asks the supplier's address server where to report. The answer is the client's server. From then on the reports go there.



No.	What happens
1	The helmet switches on and asks the address server where to report.
2	The address server answers with the two addresses we set. They point to the client's server.
3	The helmet opens a live link to the live server and logs in with its device number.
4	It sends a report every 10 seconds with its position and battery and the Bluetooth devices it hears.
5	If an alarm happens it sends the alarm at once.
6	The data store keeps every report and the screens show it.

03 The data we get from the helmet

The helmet keeps a live link open and sends its data to us. We do not have to ask for it. A report arrives every 10 seconds.

What the helmet sends	What it tells us	Where we use it in v1.00
Position (x_point and y_point)	Where the worker is. The system is Gaode (question 3).	Overview and one worker and alarm
Position quality (c_accuracy and gps and gps_level)	How far to trust the position. Whether it comes from GPS or the mobile network.	Map and last seen
Battery (bat_l and charging)	Battery percent and whether it is charging.	Overview low battery and sign in
Worn or not (is_weared)	Whether the helmet is on the head.	Sign in and shift view
Network (net_type and net_strenth)	4G or Wi-Fi and how strong.	One worker
Bluetooth devices (bt_signal_info)	Every device the helmet hears with signal strength and time.	Bluetooth test
Alarms (ca_sos)	Which alarm: SOS and fall and helmet off and others, with the position.	Alarm screen and overview
SOS button (ca_sip_sos)	The worker pressed SOS. It also asks for a call room.	Alarm screen

04 What we send back to the helmet

What we send	What it does	When it reaches the helmet
Replies to login and to each report	The helmet needs these to carry on.	At once
Settings in the start-up reply	Alarm temperature and helmet off delay and no movement time and beacon scanning and others.	When the helmet next starts up
Start or stop recording	Starts or stops the video recording on the helmet.	At once
Join a Wi-Fi network	Tells the helmet which Wi-Fi to join.	At once

HOSTING

v1.00 is built and tested on a 90Ten test server with sample data. It is then installed on the client's own servers in South Africa and the data stays there.

05 A shift from start to end

What happens in v1.00 during one shift for one worker.

When	What happens	Where
Before the shift	At the lamp room the worker is linked to a helmet and a shift. The screen shows if the helmet is seen and its battery.	Sign in screen
Shift starts	The helmet is switched on and asks where to report and logs in to the client server.	Helmet and live server
During the shift	The helmet reports every 10 seconds. The control room sees the worker on the map and who is seen and who is not.	Overview and shift view
An alarm	A fall or an SOS reaches the server at once. The control room sees who and where and when. A video call needs a call server so it comes in a later version.	Alarm screen
A headcount	The control room lists everyone signed in by where they were last seen and ticks people safe.	Headcount screen
Shift ends	The worker hands the helmet back and it is returned in the sign in screen. The history keeps who had it.	Sign in screen

NOT IN V1.00

The underground trail and data saved with no signal and fences and live video and video calls. They come in later versions.

06 What we depend on

What	Why	Status
A sample helmet	Imperium has two helmets in South Africa. We need one in hand to build the application because we have to test it. Without one we can only test with a pretend helmet.	Needed first
The admin login on the supplier platform	It is how we change the two addresses on the supplier platform.	Needed
The supplier address server being reachable	The helmet asks it first every time it starts. If it cannot be reached the helmet may not learn where to report. Changing the address inside the helmet later removes this.	To confirm
Secure connections	The addresses in the example are plain http and ws. We need to know if the helmet accepts secure addresses.	To confirm
The SIM card data	The helmet reports over 4G. Someone pays for the data.	To confirm
The map	Positions use the Gaode system (question 3). We test with a helmet.	To confirm

07 What could go wrong and what we do

What	What we do	When
The supplier address server is down	The helmet may not report. This is the reason to move to an address server of the client.	Before wide use
A helmet is not in the allowed list	The live server refuses it. It shows in the shift view as a helmet nobody signed for.	v1.00
A helmet has no signal	It shows as not seen with the last place and time. On the surface this is a prompt to check.	v1.00
Underground there is no signal	We cannot see the worker. Data saved with no signal is not described yet.	v2.00